

Math 1232 Summer 2025

Mastery Quiz 10

Due Monday, August 4

This mastery quiz has three topics. Everyone should do M4, but if you already have a 4/4 on M3 or a 2/2 on S7, feel free to skip these topics. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write “yes” or “no” or give a single number.

Please turn in this quiz in class on Monday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 3: Series Convergence
- Major Topic 4: Taylor Series
- Secondary Topic 7: Power Series

Name: Solutions

Major Topic 3: Series Convergence

Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

- (a) Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{n^2 + n - 3}{n^2 4^n}$.

Solution. We use the ratio test. We have

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)^2 + n + 1 - 3}{(n+1)^2 4^{n+1}}}{\frac{n^2 + n - 3}{n^2 4^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n^2 + 2n - 1)n^2 4^n}{(n+1)^2 4^{n+1} (n^2 + n - 3)} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{n^4 + 2n^3 - n^2}{4n^4 + 12n^3 - 20n - 15} \right| = \frac{1}{4} < 1. \end{aligned}$$

So by the ratio test this converges absolutely.

- (b) Analyze the convergence of the series $\sum_{n=1}^{\infty} n e^{-n^2+1}$.

Solution. We can work this out with the integral test. We have

$$\int_1^{\infty} x e^{-x^2+1} dx = \lim_{t \rightarrow \infty} \int_1^t x e^{-x^2+1} dx = \lim_{t \rightarrow \infty} \left. \frac{-1}{2} e^{-x^2+1} \right|_1^t = \lim_{t \rightarrow \infty} \frac{1}{2} e^0 - \frac{1}{2} e^{-t^2+1} = 1/2 < \infty.$$

Since this integral converges, the series must also converge by the integral test. Since the series is all positive, it converges absolutely.

Alternatively, we could use the ratio test. We have

$$\lim_{n \rightarrow \infty} \frac{(n+1)e^{-n^2-2n}}{n e^{-n^2+1}} = \lim_{n \rightarrow \infty} \frac{n+1}{n} \frac{e^{n^2-1}}{e^{n^2+2n}} = \lim_{n \rightarrow \infty} \frac{n+1}{n} \frac{1}{e^{2n+1}} = 0.$$

Since $0 < 1$, this converges absolutely by the Ratio Test.

- (c) Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{n! + 1}{(n+1)!}$.

Solution. We compute that

$$\frac{n! + 1}{(n+1)!} \geq \frac{n!}{(n+1)!} = \frac{1}{n+1}$$

and $\sum_{n=1}^{\infty} \frac{1}{n+1}$ diverges, so this series also diverges by the Direct Comparison Test.

Major Topic 4: Taylor Series

(a) In class we computed a Taylor series for $\sin(x)$ centered at zero. Use the degree-seven Taylor polynomial to approximate $\sin(3) \approx T_7(3, 0)$. (You don't need to numerically simplify this.) Using the Taylor series remainder, find an upper bound for the error in this approximation.

Solution. We know that

$$\begin{aligned}\sin(x) &= \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\ T_7(x, 0) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} \\ T_7(x, 3) &= 3 - \frac{27}{3!} + \frac{3^5}{5!} - \frac{3^7}{7!} = 3 - \frac{37}{6} + \frac{243}{120} - \frac{2187}{5040} \\ &= 3 - \frac{9}{2} + \frac{81}{40} - \frac{243}{560} = \frac{51}{560} \approx 0.091.\end{aligned}$$

We know that $f^{n+1}(x) = \pm \cos(x)$ or $\pm \sin(x)$ so $|f^{n+1}(z)| \leq 1$, and thus

$$\begin{aligned}|R_n(x)| &= \left| \frac{f^{(n+1)}(z)}{(n+1)!} x^{n+1} \right| \leq \frac{|x|^{n+1}}{(n+1)!} \\ |R_7(x)| &\leq \frac{|x|^{7+1}}{(7+1)!} \\ |R_7(3)| &\leq \frac{3^8}{8!} = \frac{729}{4480} \approx 0.16.\end{aligned}$$

It would also be okay to observe that the eighth term is zero, so we could actually compute

$$\begin{aligned}|R_7(x)| &= |R_8(x)| \leq \frac{|x|^{8+1}}{(8+1)!} \\ |R_8(3)| &\leq \frac{3^9}{9!} = \frac{243}{4480} \approx 0.054.\end{aligned}$$

Name: [Solutions](#)

(b) Using series we already know, write down a formula for the (infinite) Taylor series for $x^3 e^{(x^5/4)}$, and then write down the first four non-zero terms of this series.

Solution.

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{1}{n!} x^n \\ e^{x^5/4} &= \sum_{n=0}^{\infty} \frac{1}{n!} (x^5/4)^n = \sum_{n=0}^{\infty} \frac{1}{n! \cdot 4^n} x^{5n} \\ x^3 e^{x^5/4} &= \sum_{n=0}^{\infty} \frac{1}{n! \cdot 4^n} x^{5n+3} \end{aligned}$$

The first four non-zero terms are

$$x^3 + \frac{1}{4}x^8 + \frac{1}{32}x^{13} + \frac{1}{6 \cdot 64}x^{18}.$$

(Note: this is not T_3 or T_4 . It's T_{18} !)

(c) If $f(x) = \sum_{n=0}^{\infty} 2^n n^3 (x-2)^n$, compute $\frac{d}{dx} f(x)$ and $\int f(x) dx$.

Solution.

$$\begin{aligned} \frac{d}{dx} f(x) &= \sum_{n=0}^{\infty} 2^n n^4 (x-2)^{n-1} \\ (\text{or better}) &= \sum_{n=1}^{\infty} 2^n n^4 (x-2)^{n-1} \\ \int f(x) dx &= \sum_{n=0}^{\infty} \frac{2^n n^3}{n+1} (x-2)^{n+1} + C \\ (\text{or}) &= \sum_{n=1}^{\infty} \frac{2^{n-1} (n-1)^3}{n} (x-2)^n + C. \end{aligned}$$

Secondary Topic 7: Power Series

- (a) Find the radius of convergence and the interval of convergence of $\sum_{n=0}^{\infty} (n(x-3))^n$.

Solution. We use the ratio test.

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)^{n+1}(x-3)^{n+1}}{n^n(x-3)^n} \right| = \lim_{n \rightarrow \infty} |x-3| \left(\frac{n+1}{n} \right)^n (n+1) \geq |x-3| \lim_{n \rightarrow \infty} (n+1)$$

and this is infinite unless $x = 3$. So the radius of convergence is 0, and the series converges if and only if $x = 3$.

- (b) Find the radius of convergence and the interval of convergence of $\sum_{n=1}^{\infty} \frac{3^n(x+2)^n}{\ln(3n)}$.

Solution. We use the ratio test.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{\frac{3^{n+1}(x+2)^{n+1}}{\ln(3n+3)}}{\frac{3^n(x+2)^n}{\ln(3n)}} \right| &= \lim_{n \rightarrow \infty} 3|x+2| \frac{\ln(3n+3)}{\ln(3n)} \\ &= 3|x+2|. \end{aligned}$$

So we need $3|x+2| < 1$ or $-1 < 3x+6 < 1$, or $-7/3 < x < -5/3$. We need to have x in the interval $(-7/3, -5/3)$, so the radius is $1/3$.

To find the interval we need to check the endpoints. We see

$$\sum_{n=0}^{\infty} \frac{3^n}{\ln(3n)} (1/3)^n = \sum_{n=0}^{\infty} \frac{1}{\ln(3n)}$$

diverges by comparison to the harmonic series

(for example $\lim_{n \rightarrow \infty} \frac{\frac{1}{\ln(3n)}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{\ln(3n)} = \infty$, and since we've

compared to a divergent series, the LCT gives divergence)

$$\sum_{n=0}^{\infty} \frac{3^n}{\ln(3n)} (-1/3)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{\ln(3n)}$$

converges by the alternating series test

Thus the interval of convergence is $[-7/3, -5/3)$.